

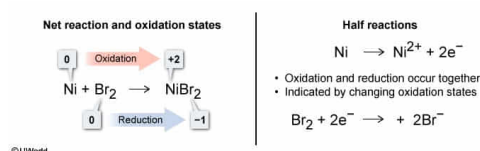
Shiny nickel metal granules added to orange liquid bromine mixed in alcohol produced a combination reaction yielding a blue-green solution of nickel(II) bromide. In the reaction between Ni and Br₂:

- ☐ A. bromine forms a chelate with nickel. [14%]
- ✓ ☒ B. nickel undergoes an oxidation-reduction reaction with bromine. [67%]
- ☐ C. the reaction forms a precipitate of nickel(II) bromide. [12%]
- ☐ D. nickel and bromine participate in a Brønsted-Lowry acid-base neutralization. [5%]

Correct

67% answered correctly

Explanation:



Many reactions can be classified as one of **five common types**, but they can also be further classified based on the results of the reaction. For example, reactions producing an insoluble solid (precipitate) crashing from the solution are said to be **precipitate formation** reactions whereas those involving **Brønsted-Lowry** acids and bases that react to produce an ionic salt and water are classified as **acid-base neutralization** reactions.

Two other important reaction classifications are chelate formation and redox reactions. In **chelate formation** reactions, a metal cation and a ligand react to form one or more rings via a pincer-like **coordinate bonding** arrangement. In an **oxidation-reduction (redox) reaction**, the oxidation states of some atoms change during the conversion of the reactants to the products.

The given reaction between Ni and Br₂ is a **combination reaction** in which two reactants form a single ionic product, NiBr₂. The assigned **oxidation states** for the elements in this reaction show that nickel goes from an oxidation state of zero in its elemental metal form to a state of +2 in the product (an **oxidation**), and bromine goes from an oxidation state of zero to a state of -1 (a **reduction**). Therefore, in addition to being a combination reaction, the reaction between Ni and Br₂ is also an oxidation-reduction reaction.

(Choice A) The reaction is not a chelate formation reaction because nickel(II) bromide is an ionic compound and does not have a ringed structure formed from pincer-like coordinate bonds between the metal and a ligand.

(Choice C) The reaction yields a blue-green *solution* of nickel(II) bromide rather than an insoluble solid precipitate.

(Choice D) Neither Ni nor Br₂ are a Brønsted-Lowry acid or base.

Educational objective:

Many reactions can be classified as one of five common types, but they can also be further classified based on the results of the reaction. Classifications include precipitate formation, chelate formation, and oxidation-reduction reactions.

Time spent: 0 sec

Subject:
General Chemistry

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Interactions of Chemical Substances